

T-Junction Traffic Light Controller using Verilog HDL

1. Introduction

Traffic light controllers play an essential role in managing vehicle movement and ensuring road safety at intersections. This project implements a T-Junction Traffic Light Controller using Verilog HDL. The controller manages signal sequencing for a three-road intersection using a timing-based control mechanism.

The design uses a counter-driven approach to activate different traffic signal groups during predefined time intervals. The project was simulated and verified using EDA Playground and EPWave.

2. Objective

The objective of this project is to design and simulate a digital traffic light controller capable of:

- Managing traffic flow at a T-junction
 - Providing safe signal transitions
 - Controlling multiple traffic movements
 - Implementing timing-based traffic control using Verilog HDL
 - Verifying functionality through simulation
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3. Tools Used

- Verilog HDL
 - EDA Playground
 - EPWave Waveform Viewer
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4. System Description

The traffic controller operates using a clock-driven counter. Based on the counter value, different traffic signal outputs are activated.

The controller divides operation into three phases:

- Phase 1: Main road traffic movement
- Phase 2: Right-turn traffic movement
- Phase 3: Side-road traffic movement

After completing all phases, the counter resets and the sequence repeats continuously.

5. Design Methodology

Step 1: Counter Initialization

A 5-bit counter is initialized and incremented on every positive clock edge.

Step 2: Signal Reset

At the beginning of each cycle, all traffic signals are cleared to prevent conflicting outputs.

Step 3: Phase Selection

The controller checks the current counter value and determines the active traffic phase.

Step 4: Signal Activation

Corresponding traffic signals are enabled according to the selected phase.

Step 5: Cycle Repetition

Once the maximum count value is reached, the counter resets and the sequence starts again.

6. Traffic Signal Phases

Phase 1 (Count 0–6)

Active Signals:

- S0
- M0
- M1
- M1_L

Purpose:

Allows traffic flow on the main road along with left-turn movement.

Phase 2 (Count 7–11)

Active Signals:

- S1
- M0
- M0_R

Purpose:

Provides right-turn access while maintaining controlled traffic movement.

Phase 3 (Count 12–16)

Active Signals:

- S2
- S

Purpose:

Allows traffic movement from the side road of the T-junction.

7. Functional Blocks

Clock Input

Provides synchronization for the controller.

Reset Logic

Initializes all outputs and counter values.

Timing Counter

Tracks the current phase of operation.

Control Logic

Determines which signals should be activated based on the counter value.

Output Signals

Generate the required traffic light control signals.

8. Simulation and Verification

The design was simulated using EDA Playground.

Verification included:

- Counter operation
- Correct phase transitions
- Proper signal activation
- Reset functionality
- Continuous cyclic operation

Waveform analysis was performed using EPWave to confirm correct behavior.

9. Results

The simulation demonstrated:

- Successful timing-based traffic control
- Correct signal sequencing
- Reliable phase transitions
- Proper reset operation

The generated waveforms confirmed that the controller behaved according to the intended traffic management strategy.

10. Applications

- Traffic signal control systems
 - Smart transportation infrastructure
 - FPGA-based traffic controllers
 - Embedded control systems
 - Educational digital design projects
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11. Learning Outcomes

Through this project, the following concepts were explored:

- Verilog HDL programming
 - Sequential logic design
 - Counter implementation
 - Timing-based control systems
 - Hardware simulation and verification
 - Digital system design
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12. Future Improvements

- Finite State Machine (FSM) implementation
 - Four-way intersection support
 - Pedestrian crossing signals
 - Vehicle density-based control
 - Emergency vehicle priority
 - FPGA hardware deployment
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13. Conclusion

The T-Junction Traffic Light Controller was successfully designed and simulated using Verilog HDL. The project demonstrates the application of timing-based digital control systems for traffic management. Through simulation and verification, the controller was shown to manage traffic flow effectively while maintaining safe and organized signal sequencing.